Cooperative Collision Avoidance for Autonomous Vehicles using Horizontal Manoeuvres

Reference	: JAAE1
Supervisor	: Dr Japie Engelbrecht
Can be completed in first semester	: No
Students allowed for this topic	: 1
Student already assigned	: None
Direction	: Mechatronic
Required elective	: None

Description:
A cooperative collision prediction and avoidance system for multiple autonomous vehicles must be designed and verified in simulation. The collision avoidance must be performed using horizontal manoeuvres only. The vehicles are assumed to have a minimum forward speed, and a maximum turn rate. It is assumed that all vehicles broadcast their current position, velocity and intent to one another. The vehicles must predict imminent collisions and then cooperatively re-plan their paths to avoid the collisions, while minimising both their control effort and their deviation from their original paths, and while obeying their own vehicle motion constraints. If time allows, the system must also be practically demonstrated using physical mobile robots.

Key modules (Scientific and Engineering Knowledge):
Control Systems (354) | Design (E) (314) | Modelling (334)